Designing Water Policy for an Interconnected World

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Introduction

Rapid economic & population growth → increased pressure on water resources

- Globalization → water-intense specialization (agriculture) in Brazil, China, India, US, ...
- Excessive water use (w.r.t. to recharge rates) → billions exposed to future water insecurity

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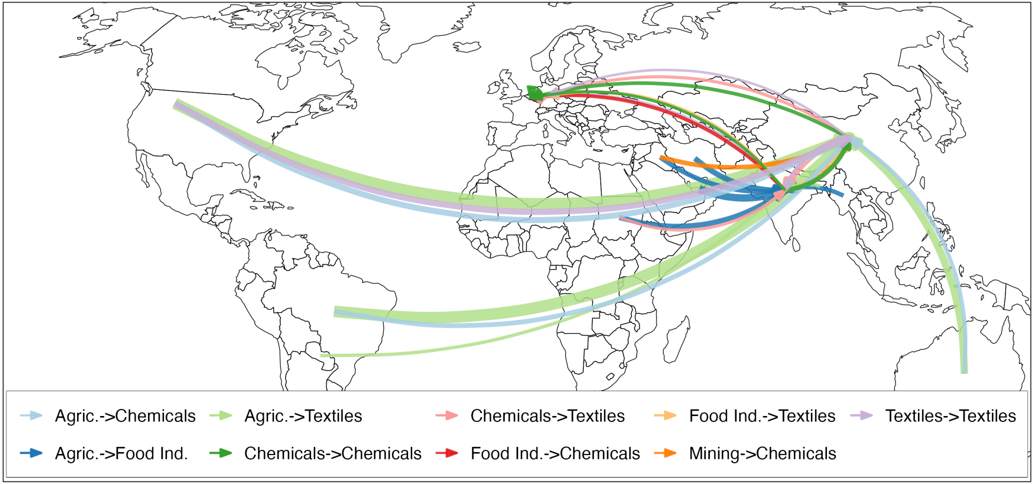
This paper – these **local challenges reflect a global problem** due to spatial linkages

Globalized Economy → Globalized Water Footprint...

Fig. Water footprint (water m³/USD) of Germany's textile industry (GLORIA data, 2020)



... Amplified by the Global Water Cycle

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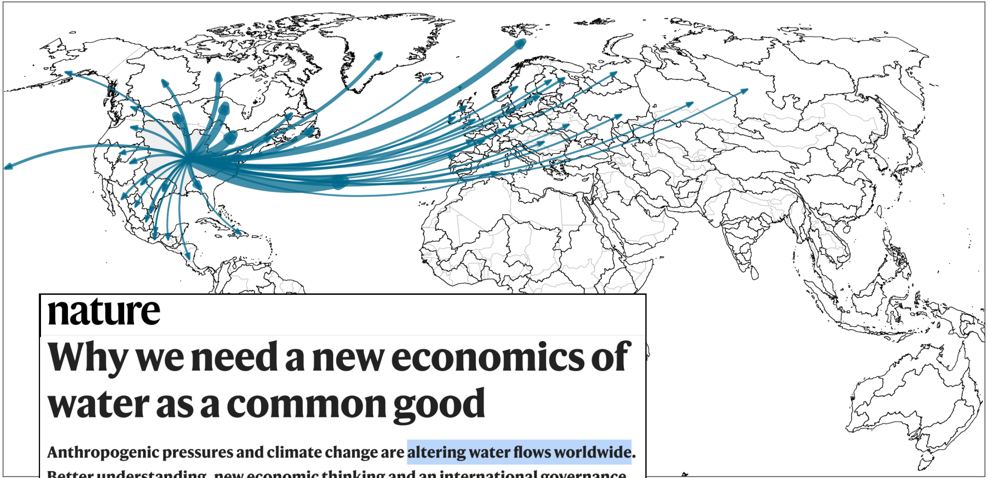

nature

Why we need a new economics of water as a common good

Anthropogenic pressures and climate change are altering water flows worldwide. Better understanding, new economic thinking and an international governance framework are needed to stave off catastrophe.

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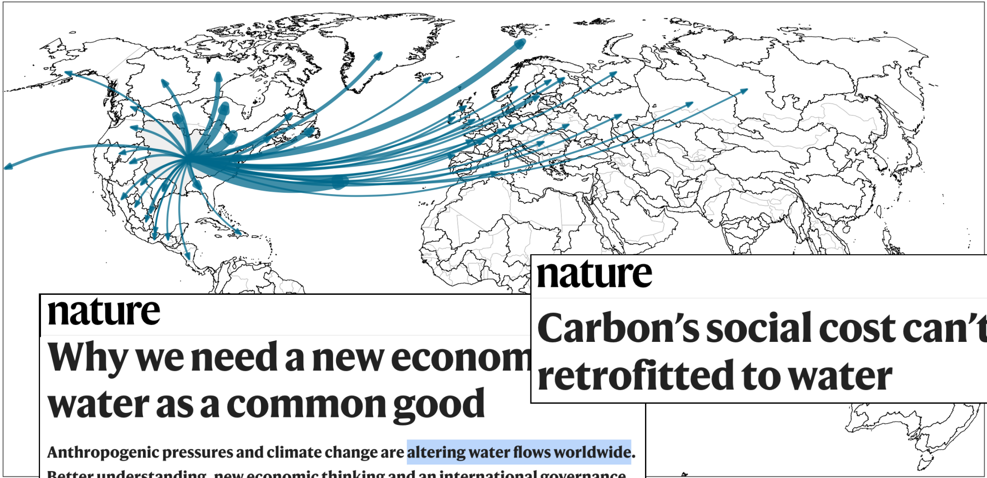

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Carbon's social cost can't be retrofitted to water

What We Do (so far)

How does the co-evolution of economy & water cycle $\rightarrow$ water security and welfare?

- *Can policies correct all (spatial and intertemporal) unpriced externalities?*

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2. Develop a dynamic spatial model <u>integrated to the global water cycle</u>
   - Common pool resource externalities $\rightarrow$ **globalized tragedy of the commons**

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2. Develop a dynamic spatial model integrated to the global water cycle
   - Common pool resource externalities $\rightarrow$ **globalized tragedy of the commons**
3. Derive an optimal water policy rule to internalize all unpriced externalities
   - Pigouvian price on water $\leftrightarrow$ spatial "centrality" of water resources

Related Literature policy contr.

Spatial-climate economics study the spatial climate-economy nexus – trade, taxes, policies (Nordhaus, 1994; Desmet and Rossi-Hansberg, 2024) more

- *Focus on water: the first integrated assessment model of the **global water cycle***

Hydrology vs. water economics (e.g., Niazi et al., 2024; Carleton et al., 2025): rich models that yet lack of full integration (or context specific) water econ.

- *Bridge both fields with a fully integrated framework **for policy evaluation***

Resource economics on the "limits to growth" & resource content of trade (Hotelling, 1931; Vanek, 1959; Solow, 1974; Davis and Weinstein, 2001) more

- *Study the intertemporal exploitation of **spatially interconnected resources***

Roadmap

1. Data & Motivating Facts
2. A Dynamic Spatial Model of the Global Economy & Water Cycle
   - Integrated Model and Stylised Simulations
   - Optimal Water Policy for an Interconnected World
3. Next Steps: Linking Theory to Data + Counterfactuals + ...?

Motivating facts: How interconnected are the global water cycle & economy?

Preamble: A Closed-System, Spatially Connected Water Cycle

Without economic water use, water stocks R in (any) geography i, j evolves as:

$$R_i(t+1) = R_i(t) - \underbrace{E_i(t)}_{\kappa_i R_i(t)} - \underbrace{S_i(t)}_{\delta_i R_i(t)} + \underbrace{\tilde{E}_i(t)}_{\sum_j f_{ji}^{(E)} \kappa_j R_j(t)} + \underbrace{\tilde{S}_i(t)}_{\sum_j f_{ji}^{(S)} \delta_j R_j(t)}$$

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Outflows (evaporation & surface runoffs)

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Recharge (precipitation & surface runoffs)

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Atmospheric linkages

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Economic water use $H_i(t)$ distorts eq. (1) – but in an ambiguous direction

- **Closed global system** $\rightarrow \sum_i R_i(t) = \sum_i R_i(t+1)\ \forall t$
- Withdraws return to water system (e.g., evotranspiration, surface runoff)

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Next: empirical facts that corroborate these mechanisms (e.g., strength of $\{f_{ji}^{(E)}\}_{j,i}$)

Fact 1: Global Atmospheric Links Between Water Resources

RECON data: **atmospheric moisture flows** (0.5° x 0.5°, De Petrillo et al., 2025a,b) details

- Evap.$\rightarrow$Precip. flows (m^3, $\sim$2008-17)

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Our focus: major land water basins (GAEZ)

- Basins $\equiv$ precipitation catchment areas (no surface runoff b/w basins)

Fig. Land Water Basins from FAO GAEZ

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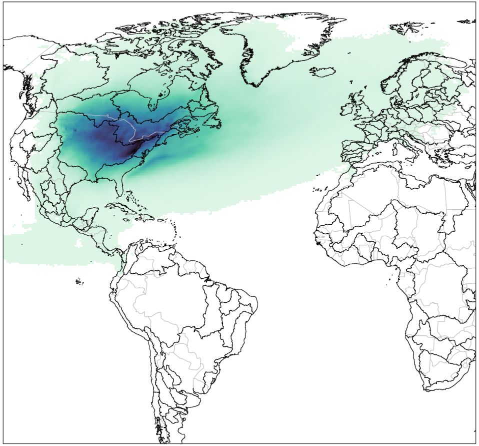

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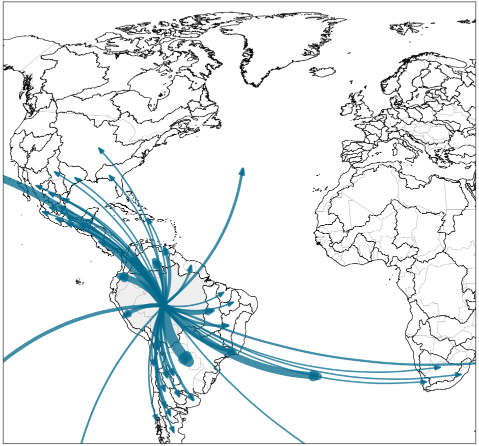

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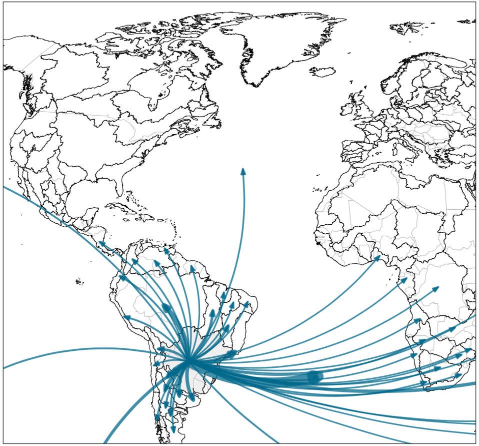

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Table 1: Water "import shares" per origin basin type (%) maps export shares

	Own basin	Other basins (domestic)	Other basins (foreign)	Oceans
Min.	0.65	0.09	0.17	5.82
Q1	2.41	3.68	7.94	52.99
Median	4.81	8.62	17.71	68.72
Mean	7.68	11.12	20.04	63.99
Q3	9.61	16.59	28.58	79.09
Max.	48.66	50.93	71.27	98.18

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Median basin $\sim$ 25% from other land basins

Fact 2: Economic Activity Explains Δ Global Water Cycle

Does economic water usage explains observed Δ water cycle? Data:

- Country-level economic water use (1960-2020; World Bank & FAO) details
- Evaporation and precipitation across space and time ERA5 data
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Country-level long-differ. (Burke and Emerick, 2016; Carleton et al., 2024) $\Delta \equiv$ trends

$$\Delta\text{Evaporation}_c = \alpha\Delta\text{Water use}_c + \beta\Delta X_c + \varepsilon_c \tag{2}$$

$$\Delta\text{Precipitation}_c = \alpha\Delta\text{Water use}_c + \beta\Delta X_c + \gamma\Delta\text{Foreign water use}_c + \varepsilon_c, \tag{3}$$

where $\Delta\text{Foreign water use}_c = \sum_{c' \neq c} f_{c'c}^{(E)} \Delta\text{Water use}_{c'}$

Fact 2: Economic Activity Explains Δ Global Water Cycle

Table 2: Regression results – global water cycle and economic water use ≠outcomes IV

	Δ Evaporation			Δ Precipitation		
	(1)	(2)	(3)	(4)	(5)	(6)
Δ Water use	0.6204***	0.6584***		0.6427***		0.5305***
	(0.0738)	(0.0741)		(0.0870)		(0.0908)
Δ Temperature		0.0329**	0.0096	0.0072	-0.0155	0.0204
		(0.0130)	(0.0156)	(0.0152)	(0.0173)	(0.0153)
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Observations	156	156	156	156	156	156
R^2	0.31439	0.34199	0.00243	0.26692	0.00519	0.31649

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Systematic relationship with Δ Water use

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ΔEvap. $\perp$ Δglobal warming

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Precipitation: similar pattern

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ΔForeign $\sim 1/3$ ΔWater!

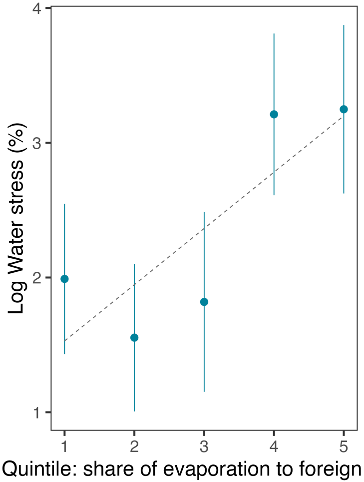
Fact 3: Most water-stressed countries $\leftrightarrow$ top "atm. exporters"

Water stress $\equiv$ water withdraws/endowments FAO data

Top water-stressed countries:

- Also the top "atmospheric water exporters"
- I.e., most central countries in terms of "propagation" of local water depletion

Fig. Water exporters vs. water stress



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GLORIA MRIO (164 countries, 46 sectors, 1995-2020)

- Water m^3/USD $\rightarrow$ water content of trade details
- 2000-20: $\uparrow$ specialization w.r.t. to traded water

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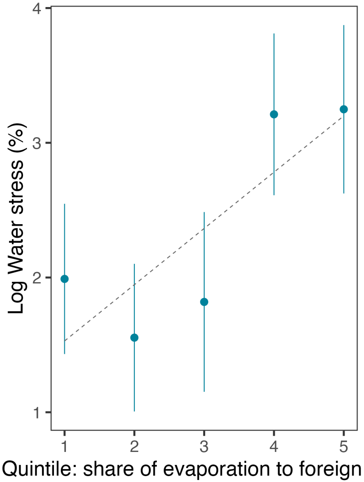

Motivating Facts: Taking Stock

Global economy & water cycle – highly interconnected networks

- Local water exploitation $\rightarrow$ global consequences (via feedback loops)
- Current water challenges: cross-sectional evidence of a slow-moving process

How will the economy & water co-evolve, and where it will lead is unknown

Next: an integrated model that incorporates such interconnections

- Optimal water policy that corrects all water-related externalities

Theory: A Dynamic Spatial Model of the Global Economy & Water Cycle

Dynamic Spatial Model – Overview

Economic module: trade model with labor and water as inputs

- Subject to policies: tariffs, subsidies, water prices
- Water use $\rightarrow$ higher extraction/prod. costs $\rightarrow$ shifts in trade & production

Water cycle module: closed system of water basins whose stock are

- Depleted by evaporation (and water use), replenished by precipitation
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Water-Economy ↻ Feedback $\rightarrow$ ***Globalized Tragedy of the Commons***

(i.e., local extraction imposes unpriced costs on the rest of the world)

Dynamic Spatial Model – Environment

Geography of N countries i, j; fixed labor $\bar{L}_i$ and endogenous water stocks $R_i(t)$

- $K + 1$ industries ($k = 0$: water extraction; $k \geq 1$: final goods), time $t = 0, 1, \ldots$

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Policy instruments create a wedge b/w producer (P) and consumer prices ($\tilde{P}$):

Import Tariff

$$\tilde{P}_{ji,k} = \frac{(1 + t_{ji,k})}{(1 + s_{j,k})(1 + x_{ji,k})} \times P_{ji,k}$$

Production Subsidy **Export Subsidy**

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Production Subsidy **Export Subsidy**

Tax revenues T_i rebated to households $\rightarrow$ balanced budget: $I_i = w_i \bar{L}_i + T_i$

- Countries can also resort to water pricing $\tau_{i,k}$ (defined later)

Dynamic Spatial Model – Demand & Supply

Demand: $W_i = \mathrm{V}_i(I_i, \tilde{\boldsymbol{P}}_i)$; Marshallian demand (Roy's identity) $C_{ji,k} = \mathrm{D}_{ji,k}(I_i, \tilde{\boldsymbol{P}}_i)$; e.g.,

$$W_i = \frac{I_i}{\tilde{P}_i}, \quad \tilde{P}_i = \prod_k \left[\sum_n b_{ni,k} \, \tilde{P}_{ni,k}^{1-\sigma_k} \right]^{\frac{\beta_{i,k}}{1-\sigma_k}}, \quad \mathrm{D}_{ji,k} = \frac{b_{ji,k} \, \tilde{P}_{ji,k}^{-\sigma_k}}{\sum_n b_{ni,k} \, \tilde{P}_{ni,k}^{1-\sigma_k}} \beta_{i,k} \, I_i \quad \text{(CD–CES)}$$

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Supply: Final goods ($k \geq 1$) with CRS production $(L_{j,k}, H_{j,k})$; producer price:

$$P_{ji,k} = \bar{d}_{ji,k} \, \mathrm{c}_{j,k}(w_j, \tilde{r}_{j,k}), \quad \text{where}$$

$\tilde{r}_{j,k} = r_j + \tau_{j,k} \equiv$ effective water price (inclusive of water pricing $\tau_{j,k}$)

Dynamic Spatial Model – Water Extraction & Factor Demands

Water extraction ($k = 0$) uses labor as $Q_{j,0} = L_{j,0} / G_j(R_j) \rightarrow$

$$r_j = w_j \, G_j(R_j)$$

where $G_j' < 0 \rightarrow$ extraction costs increase as resources deplete

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Factor demands: $L_{j,k} = \frac{\partial c_{j,k}}{\partial w_j} Q_{j,k}, H_{j,k} = \frac{\partial c_{j,k}}{\partial \tilde{r}_{j,k}} Q_{j,k}$ (Shephard's lemma)

- $H_{j,k}$ rises when R_j is abundant (low r_j) and falls as the resource depletes

Myopic agents $\rightarrow$ ***Common Pool Resource Externality*** (spatially and intertemporally)

(i.e., firms take r_j as given, not internalizing that withdrawals lower R_j and affect water stocks (and prices) in all connected basins)

Water Cycle – Basin Geography & Natural Hydrology

Global water cycle as a closed system of land basins $\mathcal{B} = \{1, \ldots, B\}$ + ocean $\mathcal{O} = \{0\}$

- Oceans treated as a stable reservoir (i.e., $E_0 = \bar{E}_0$, $S_0 = \bar{S}_0$ exogenous)

Without economic activity, water stocks evolve as

$$R_i(t+1) = R_i(t) - \underbrace{E_i(t)}_{\kappa_i R_i(t)} - \underbrace{S_i(t)}_{\delta_i R_i(t)} + \underbrace{\tilde{E}_i(t)}_{\sum_j f_{ji}^{(E)} E_j(t)} + \underbrace{\tilde{S}_i(t)}_{\sum_j f_{ji}^{(S)} R_j(t)}$$

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Stacking **land basins** $i \geq 1 \rightarrow$ law of motion:

$$\boldsymbol{R}_B(t+1) = \boldsymbol{\Gamma} + \boldsymbol{\Phi}\, \boldsymbol{R}_B(t)$$

$$\boldsymbol{\Gamma} \equiv \boldsymbol{F}_{OB}^{(E)}\, \bar{\boldsymbol{E}}_0 + \boldsymbol{F}_{OB}^{(S)}\, \bar{\boldsymbol{S}}_0 \qquad \textit{(exogenous ocean inflows)}$$

$$\boldsymbol{\Phi} \equiv \boldsymbol{I} - \boldsymbol{\kappa} - \boldsymbol{\delta} + \boldsymbol{F}_{BB}^{(E)}\, \boldsymbol{\kappa} \qquad \textit{(net retention: losses + atm. recycling)}$$

Water Cycle: Economic Withdrawals & Consumptive Use

Now add economic activity with consumptive-use share $\theta_{ik} \in (0,1)$:

- Fraction of withdrawal lost via evaporation; remainder returns as runoff

Basin-level aggregates:

$$H_i = \sum_k H_{ik}, \qquad \theta_i(t) = \frac{\sum_k \theta_{ik} H_{ik}}{\sum_k H_{ik}} \quad \text{(time-varying with sectoral mix)}$$

Modified evapotranspiration (natural + economic):

$$\boldsymbol{E}_B(t) = \boldsymbol{\kappa}\, \boldsymbol{R}_B(t) + \boldsymbol{\theta}(t)\, \boldsymbol{H}(t)$$

Economic withdrawals $\uparrow$ evapotranspiration, shifting precipitation $\forall$ basins

Water Cycle: Updated Law of Motion & Steady State

Law of motion with economic activity:

$$\boldsymbol{R}_B(t+1) = \boldsymbol{\Gamma} + \boldsymbol{\Phi}\,\boldsymbol{R}_B(t) + \boldsymbol{\Psi}(t)\,\boldsymbol{H}(t), \qquad \boldsymbol{\Psi}(t) \equiv (\boldsymbol{F}_{BB}^{(E)} - \boldsymbol{I})\,\boldsymbol{\theta}(t)$$

$\boldsymbol{\Psi}(t) \equiv$ *net economic retention*, with an ambiguous sign

Steady state with economic activity (fixed-point in $\boldsymbol{R}_B^{(ss)}$):

$$\boldsymbol{R}_B^{(ss)} = (\boldsymbol{I} - \boldsymbol{\Phi})^{-1}\Big(\boldsymbol{\Gamma} + \boldsymbol{\Psi}^{(ss)}\,\boldsymbol{H}(\boldsymbol{R}_B^{(ss)})\Big)$$

Closed system: $\sum_i R_i(t+1) = \sum_i R_i(t)$ – fixed global water stock

General Equilibrium

Static equilibrium – wages $\{w_i\}$ simultaneously clear:

$$\text{(Goods)}\; Q_{i,k} = \sum_n \bar{d}_{in,k}\, C_{in,k} \quad \text{(Water)} \quad \frac{L_{i,0}}{G_i(R_i)} = \sum_{k=1}^{K} H_{i,k} \quad \text{(Labor)}\; \bar{L}_i - L_{i,0} = \sum_{k=1}^{K} L_{i,k}$$

Dynamic equilibrium: sequence of static equilibria $\{w_i(t)\}_t$ with $R_i(0)$ given

- Stocks $\{R_i(t)\}_t$ evolving per the LOM above (state variable)

Steady-state equilibrium: dynamic equilibrium satisfying the $\boldsymbol{R}_B^{(ss)}$ fixed-point

solving algor. simulations

Optimal Water Policy for and Interconnected World

Optimal Policy – Problem & Water Stock Linearization

Optimal policy $\mathbb{I}_i \equiv \{\boldsymbol{t}_i, \boldsymbol{x}_i, \boldsymbol{s}_i, \tau_i\}$ + transfers Δ_i ($\sum_i \Delta_i = 0$) solves

$$\max_{\mathbb{I}} \ \mathcal{W}(W_1(\mathbb{I}), \ldots, W_N(\mathbb{I})) \qquad \text{s.to steady-state GE relationships}$$

Optimal Policy – Problem & Water Stock Linearization

Optimal policy $\mathbb{I}_i \equiv \{\boldsymbol{t}_i, \boldsymbol{x}_i, \boldsymbol{s}_i, \boldsymbol{\tau}_i\}$ + transfers Δ_i ($\sum_i \Delta_i = 0$) solves

$$\max_{\mathbb{I}} \ \mathcal{W}(W_1(\mathbb{I}), \ldots, W_N(\mathbb{I})) \qquad \text{s.to steady-state GE relationships}$$

Re-write steady-state water stocks (from the LOM fixed point) as:

$$\boldsymbol{R}_B^{(ss)} = \boldsymbol{Z} + \boldsymbol{\Lambda}^{(ss)} \boldsymbol{H}^{(ss)}, \qquad \boldsymbol{Z} \equiv (\boldsymbol{I} - \boldsymbol{\Phi})^{-1} \boldsymbol{\Gamma}, \quad \boldsymbol{\Lambda}^{(ss)} \equiv (\boldsymbol{I} - \boldsymbol{\Phi})^{-1} \boldsymbol{\Psi}^{(ss)}$$

$$R_i = Z_i + \sum_j \sum_k \lambda_{ji,k} H_{j,k} \quad \text{[country level]}$$

$\lambda_{ji,k}$: hydrological "impact factor" of withdrawals at (j, k) on steady-state R_i

Optimal Water Policy – Tax Rates derivation

Non-water taxes: $t_{ij,k} = x_{ij,k} = s_{i,k} = 0$

- From a global perspective, distorting goods prices is always inefficient

Optimal Water Policy – Tax Rates derivation

Non-water taxes: $t_{ij,k} = x_{ij,k} = s_{i,k} = 0$

- From a global perspective, distorting goods prices is always inefficient

Optimal water tax (Pigouvian + spatial linkages):

$$\tau_{i,k} = \tilde{r}_{i,k} - r_i = \sum_n \frac{G'_n(\cdot)}{G_n(\cdot)} r_n \sum_g \lambda_{in,g} H_{n,g}$$

- $G'_n / G_n < 0$: *extraction externality* – depletion $\rightarrow$ higher future unit costs
- $\lambda_{in,g}$: *spatial weights* – hydrological linkages propagate the externality globally

Optimal Policy – International Transfers

Optimal income shares achieving any distributional objective:

$$\pi_i^* = \frac{\frac{\partial \mathcal{W}}{\partial W_i} \frac{\partial \mathrm{V}_i}{\partial E_i} I_i}{\sum_n \frac{\partial \mathcal{W}}{\partial W_n} \frac{\partial \mathrm{V}_n}{\partial E_n} I_n}, \qquad \Delta_i^* = \pi_i^* \sum_n I_n - I_i$$

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Special case: log SWF $\mathcal{W} = \sum_i \omega_i \ln W_i$ + homothetic demand

- $\pi_i^* = \omega_i \Rightarrow$ efficient frontier traced by varying $\{\omega_i\}$

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$$\pi_i^* = \frac{\frac{\partial \mathcal{W}}{\partial W_i}\frac{\partial \mathrm{V}_i}{\partial E_i} I_i}{\sum_n \frac{\partial \mathcal{W}}{\partial W_n}\frac{\partial \mathrm{V}_n}{\partial E_n} I_n}, \qquad \Delta_i^* = \pi_i^* \sum_n I_n - I_i$$

Special case: log SWF $\mathcal{W} = \sum_i \omega_i \ln W_i$ + homothetic demand

- $\pi_i^* = \omega_i \Rightarrow$ efficient frontier traced by varying $\{\omega_i\}$

***Optimal policy separates efficiency from distribution:** water taxes correct the externality; transfers achieve the distributional goal*

Final Remarks & Next Steps

Final Remarks & Next Steps

Economic activity distorts the global water cycle through spatial linkages

- Slow-moving but empirically measurable process

A dynamic spatial model integrating both networks

- Common-pool externalities $\rightarrow$ *globalized tragedy of the commons*
- Corrected by a basin-specific Pigouvian water tax $\sim$ Social Cost of Water

Next steps: link to global data $\rightarrow$ quantify welfare costs of water externalities

- Simulate optimal policies for the 21st century
- Model extension (e.g., full-fledged dynamics, non-homothetic preferences, ...)

Thank you

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Appendix

Policy Contribution: Details

Back

Policy-relevant: addresses calls for global water governance (Mazzucato et al., 2024) and HORIZON EU's environmental research priorities

Envisioned legacy: inspire new research on

- **Social cost of water**: challenges the paradigm of an inherently local SCW (Olmstead, 2010) $\rightarrow$ first calculation of shadow prices internalizing full spatial & intertemporal externalities of local water use
- **Political economy** of shared (water) resources
- Water and the **green transition**: energy-intensive technologies, AI, ...

Spatial-Climate Economics

Back

Spatial-climate economics: spatial economy—climate (Nordhaus, 1994; Hassler and Krusell, 2012; Krusell and Smith, 2022; Desmet and Rossi-Hansberg, 2015, 2024)

- Trade (Conte et al., 2021), green subsidies (Cruz and Rossi-Hansberg, 2024), CO_2 taxes (Conte et al., 2025), policy coordination (Farrokhi et al., 2025a)
- Distributional effects (Bourany, 2025a,b; Bourany and Rosenthal-Kay, 2025)

Gap: no study has focused on water as an outcome of the economy-climate system

- *This paper:* first integrated model of the global economy & global water cycle

Water Economics Literature

Back

Hydrology: rich models projecting future water challenges, but limited flexibility for policy exercises (e.g., Niazi et al., 2024)

Water economics: mostly context-specific settings

- India (e.g., Fishman et al., 2015; Sekhri, 2014; Blakeslee et al., 2023); Pakistan (Haseeb, 2024); US (e.g., Hornbeck and Keskin, 2014; Debaere and Li, 2020)

Gap: lack of full integration (e.g., exogenous recharge; Carleton et al., 2025)

- *This paper:* global, integrated & transparent framework **bridging both fields**

Resource Economics Literature

Back

"Limits to growth": intertemporal resource depletion (Solow, 1974; Stiglitz, 1974; Pindyck, 1978); resource content of trade (Vanek, 1959; Davis and Weinstein, 2001; Debaere, 2014)

Modern revival – sustainable exploitation in spatial models:

- Forests (Farrokhi et al., 2025b), fisheries (Kang, 2024), oil (Farrokhi, 2020; Hassler et al., 2022)

Gap: the modern revival has not yet extended to water

- *This paper:* intertemporal exploitation of spatially interconnected water resources

RECON Data: Details

Back

RECON (De Petrillo et al., 2025a,b): bilateral atmospheric moisture flows between every pair of 0.5° grid cells worldwide

- **Methodology:** UTrack Lagrangian moisture tracking $\rightarrow$ Iterative Proportional Fitting (IPF) reconciliation with ERA5 reanalysis
- **Coverage:** global, $0.5^\circ \times 0.5^\circ$; average year based on **2008–2017**
- **Content:** $f_{gg'}^{(E)}$ (m^3) = atmospheric moisture evaporated at cell g and precipitated at cell g'

RECON Data: Our Application Back

Basin-level aggregation $\rightarrow$ moisture matrix $\boldsymbol{F}_{BB}^{(E)}$ used in the model's law of motion:

$$f_{ji}^{(E)} = \sum_{g \in j} \sum_{g' \in i} f_{gg'}^{(E)}$$

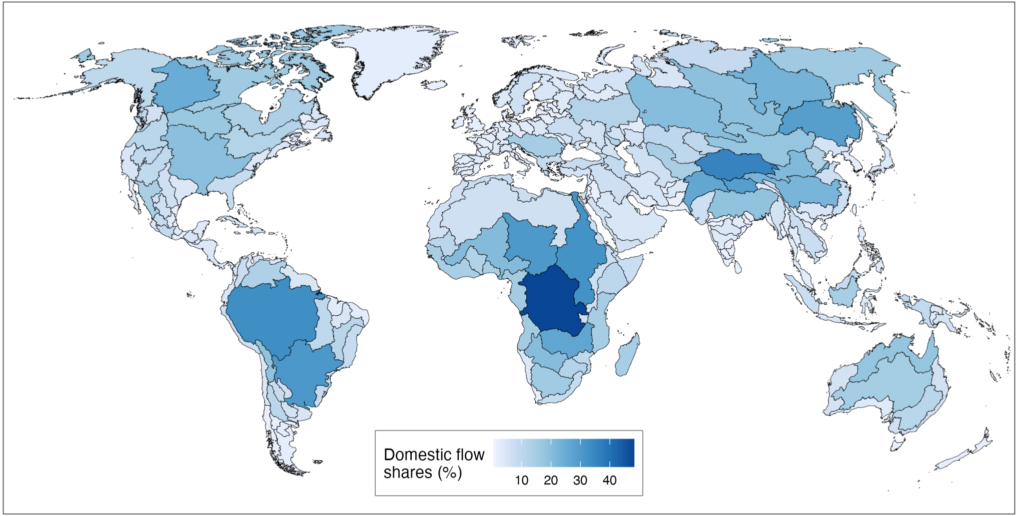
Country-level aggregation $\rightarrow$ "foreign water use" regressor in Fact 2:

$$f_{c'c}^{(E)} = \sum_{g \in c'} \sum_{g' \in c} f_{gg'}^{(E)}, \qquad \Delta\text{Foreign water use}_c = \sum_{c' \neq c} f_{c'c}^{(E)} \, \Delta\text{Water use}_{c'}$$

Atmospheric Moisture Shares: Maps

Back

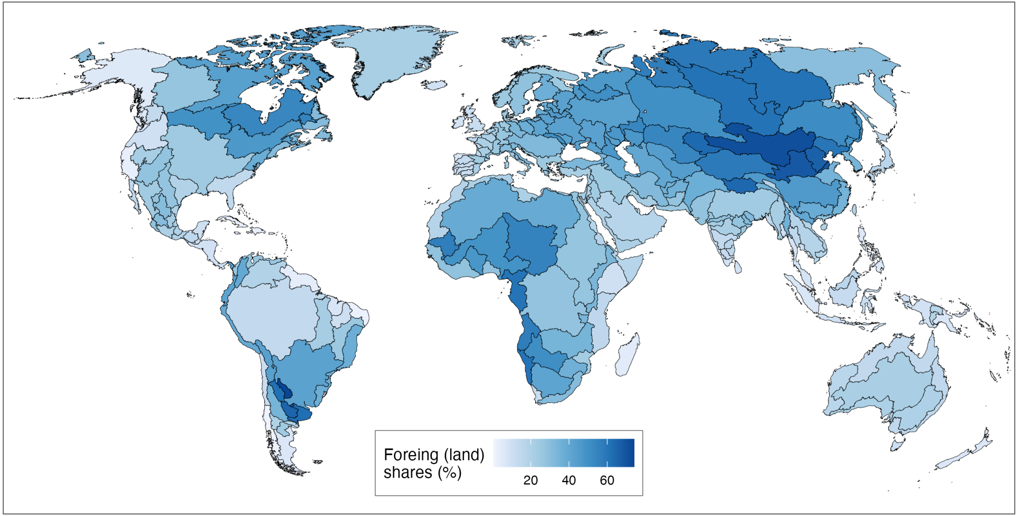
Fig. Share of atmospheric "domestic" flows



Atmospheric Moisture Shares: Maps

Back

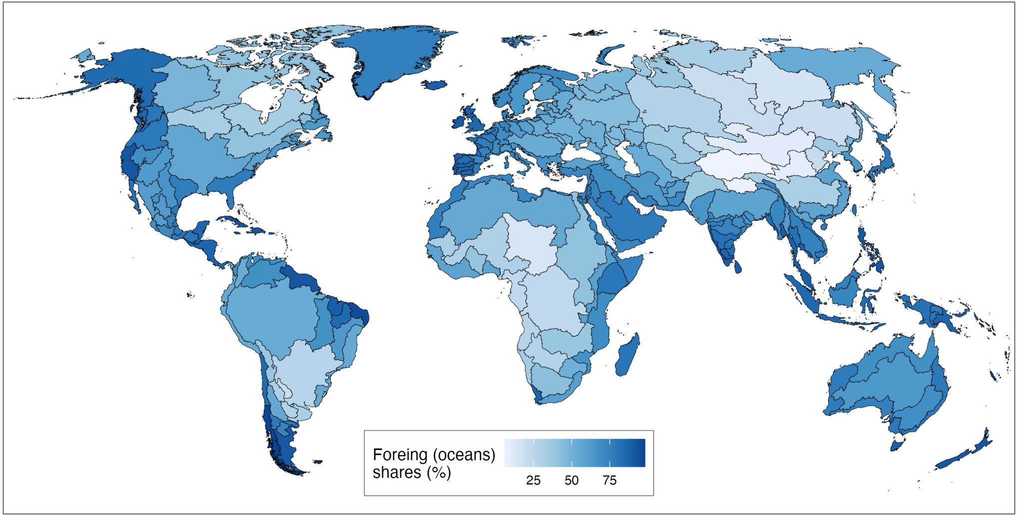
Fig. Share of atmospheric "foreign flows" from other land basins



Atmospheric Moisture Shares: Maps

Back

Fig. Share of atmospheric "foreign flows" from the oceans



Atmospheric Moisture: Export Shares

Back

Table 3: Water "export shares" per destination basin type (%)

	Own basin	Other basins (domestic)	Other basins (foreign)	Oceans
Min.	0.77	0.06	1.39	1.60
Q1	3.01	5.59	13.63	19.43
Median	6.59	13.34	28.14	34.34
Mean	11.04	20.37	36.20	37.58
Q3	13.04	32.62	57.22	53.91
Max.	74.58	80.21	91.43	94.26

ERA5 Data

Back

ERA5 (ECMWF Reanalysis v5): accessed via Copernicus Climate Data Store

- **Variables:** monthly evaporation, precipitation, and 2m temperature
- **Spatial resolution:** $0.25^\circ \times 0.25^\circ$ ($\sim$31 km at equator); global
- **Temporal span:** 1940 to present (monthly averages)

Processing: aggregate from grid-cell $\times$ month $\rightarrow$ country $\times$ year

- Area-weighted average of 0.25° cells within each country's borders
- Sum (evaporation, precipitation) or average (temperature) over months within year

World Bank / FAO Water Data

Back

FAO-AQUASTAT – main water use variable:

- Country-level freshwater withdrawals by sector (agricultural, municipal, industrial)
- 193 countries, ∼1960–present (irregular frequency by country)

World Bank Development Indicators – proxy variables:

- Agricultural value added (constant USD; % of GDP)
- Agricultural land area (km^2; % of total land area)
- ∼200 countries, 1960–present

Country Trends: Construction

Back

Measuring long-run Δ: country-level trends estimated from the panel (not endpoint differences)

- Less sensitive to year-specific outliers and business-cycle fluctuations

For each outcome o (e.g., water use), estimate country-specific linear time trends:

$$\ln y_{ct}^{o} = \sum_{c'} \gamma_{c}^{o} \cdot \left(\mathbf{1}\{c = c'\} \times \text{Year}_t\right) + \varepsilon_{ct}$$

$\hat{\gamma}_c^o \equiv$ estimated average annual growth rate (%) of outcome o in country c

- $\hat{\gamma}_c$ used as Δy_c^o in the cross-country regressions

Fact 2: Economic Activity Explains Δ Global Water Cycle

Back

Table 4: Regression results – global water cycle and agricultural VA

	Δ Evaporation			Δ Precipitation		
	(1)	(2)	(3)	(4)	(5)	(6)
Δ Agric. VA	0.6629***	0.6820***		0.6975***		0.6085***
	(0.0582)	(0.0578)		(0.0685)		(0.0701)
Δ Temperature		0.0278**	0.0096	0.0032	-0.0155	0.0147
		(0.0114)	(0.0156)	(0.0135)	(0.0173)	(0.0134)
Δ Foreign agric. VA						0.0143***
						(0.0039)
Observations	156	156	156	156	156	156
R^2	0.45708	0.47731	0.00243	0.40703	0.00519	0.45569

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Fact 2: Economic Activity Explains Δ Global Water Cycle

Back

Table 5: Regression results – global water cycle and agricultural land use

	Δ Evaporation			Δ Precipitation		
	(1)	(2)	(3)	(4)	(5)	(6)
Δ Agric. land	0.6886***	0.6955***		0.6828***		0.6210***
	(0.0388)	(0.0383)		(0.0515)		(0.0545)
Δ Temperature		0.0216**	0.0096	-0.0038	-0.0155	0.0051
		(0.0088)	(0.0156)	(0.0119)	(0.0173)	(0.0120)
Δ Foreign agric. land						0.0208***
						(0.0071)
Observations	156	156	156	156	156	156
R^2	0.67204	0.68432	0.00243	0.53678	0.00519	0.56158

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Fact 2: IV Robustness

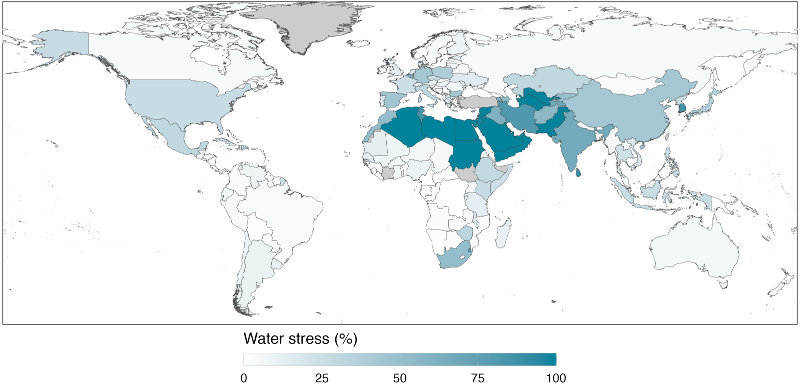
Back

Water Stress Data: FAO

Back

Definition (FAO AQUASTAT): Water stress $= \dfrac{\text{Total freshwater withdrawn}}{\text{Total renewable freshwater resources}} \times 100$

Fig. Water stress (%) across countries (FAO AQUASTAT, avg. 2008–2017)



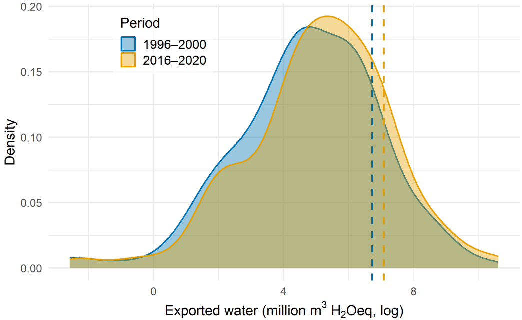
Note: Values above 100% winsorized. Grey = missing data.

GLORIA MRIO: Details Back

Data: GLORIA MRIO – 164 countries, 46 sectors, 1995–2020

Exported water content: water embedded in goods sold to foreign countries

- For each country i, sector k: water intensity ω_{ik} (m^3/USD) from GLORIA satellite accounts
- Exported water $\equiv \sum_{j \neq i} \sum_k \omega_{ik} \cdot X_{ij,k}$, where $X_{ij,k}$ = exports of k from i to j



Solving Algorithm: Static Equilibrium

Back

Iterative algorithm for static GE given water stocks $\{R_i\}$ and tax rates $\{\tau_{i,k}\}$:

1. **Initialize:** guess $\{w_i, I_i\}$; normalize wages to numeraire w_{i_0}
2. **Prices:** compute $r_i = w_i G_i(R_i)$; unit costs $c_{i,k}$; cost shares $\alpha_{i,k}$; producer and consumer prices $P_{ji,k}$, $\tilde{P}_{ji,k}$
3. **Quantities:** compute $C_{ji,k}$; revenues $Y_{i,k}$; water use $H_{i,k} = (1 - \alpha_{i,k})Y_{i,k}/(r_i + \tau_{i,k})$; water labor $L_{i,0} = G_i(R_i) \sum_k H_{i,k}$; tax revenues T_i
4. **Update & check:** $w_i^{(new)} = \frac{\sum_k \alpha_{i,k} Y_{i,k}}{\bar{L}_i - L_{i,0}}$ (Labor MCC); $I_i^{(new)} = w_i \bar{L}_i + T_i$

   $\rightarrow$ Stop if $\max_i |(w_i^{(new)} - w_i)/w_i|$ and $\max_i |(I_i^{(new)} - I_i)/I_i|$ sufficiently small; else dampen and return to Step 1

Solving Algorithm: Dynamic & Steady State

Back

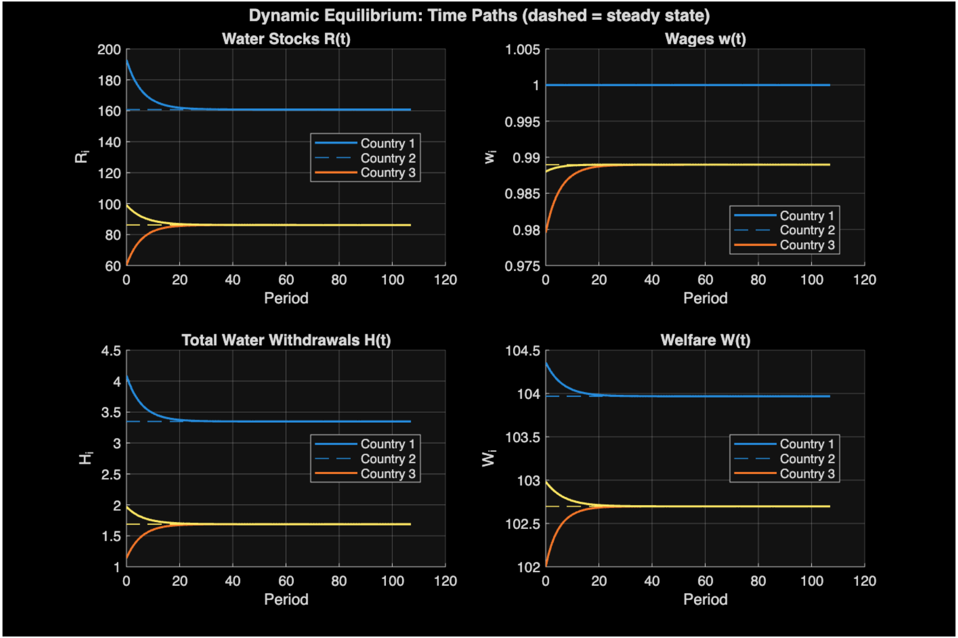
Dynamic equilibrium: solve static GE each period t; update stocks via the LOM:

$$\boldsymbol{R}_B(t+1) = \boldsymbol{\Gamma} + \boldsymbol{\Phi}\,\boldsymbol{R}_B(t) + \boldsymbol{\Psi}(t)\,\boldsymbol{H}(t)$$

Steady-state equilibrium: fixed-point intersection of two mappings

- **(a)** Static GE $\rightarrow$ $\boldsymbol{H}$ as a function of $\boldsymbol{R}_B^{(ss)}$
- **(b)** LOM fixed point: $\boldsymbol{R}_B^{(ss)} = (\boldsymbol{I} - \boldsymbol{\Phi})^{-1}\Big(\boldsymbol{\Gamma} + \boldsymbol{\Psi}^{(ss)}\boldsymbol{H}(\boldsymbol{R}_B^{(ss)})\Big)$

$\Rightarrow$ Iterate on $\boldsymbol{R}_B^{(ss)}$ until both mappings are mutually consistent



Dynamic Equilibrium: Time Paths (dashed = steady state)
Water Stocks R(t)
Wages w(t)
Total Water Withdrawals H(t)
Welfare W(t)
Country 1
Country 2
Country 3
Period

back

Optimal Policy: Dual Problem Back

Dual reformulation: planner directly chooses consumer prices
$\mathbb{P}_i \equiv \{\tilde{P}_{ii,k}, \tilde{P}_{in,k}, \tilde{P}_{ni,k}; \tilde{r}_{i,k}\}_{n \neq i,k}$

Income function for country i:

$$I_i(\cdot) = w_i L_i + \underbrace{(\tilde{\boldsymbol{r}}_i - \boldsymbol{r}_i)^\top \boldsymbol{H}_i}_{\text{water taxes}} + \underbrace{(\tilde{\boldsymbol{P}}_i - \boldsymbol{P}_i)^\top \boldsymbol{C}_i}_{\text{tariffs + prod. taxes}} + \underbrace{(\tilde{\boldsymbol{P}}_{-i} - \boldsymbol{P}_{-i})^\top \boldsymbol{C}_{-i}}_{\text{export taxes}}$$

Key functions in the dual:

- Producer price: $\mathrm{P}_{ij,k}(w_i, \tilde{r}_{i,k}) = \bar{d}_{ij,k}\, \mathrm{c}_{i,k}(w_i, \tilde{r}_{i,k})$
- Water cost: $r_i(w_i, R_i) = w_i\, \mathrm{G}_i(R_i)$
- Steady-state stock: $R_i = Z_i + \sum_{j,k} \lambda_{ji,k} H_{j,k}$

Optimal Policy: Wage Neutrality Back

Lemma 1 (Local wage neutrality): $\dfrac{\partial\,\mathrm{I}_i(\cdot)}{\partial w_i} = 0$

Proof: expand using Shephard's lemma + labor market clearing:

$$\bar{L}_i - \underbrace{\mathrm{G}_i(R_i)\sum_k H_{i,k}}_{=L_{i,0}} - \underbrace{\sum_k \frac{\alpha_{i,k} P_{ii,k} Q_{i,k}}{w_i}}_{=\sum_k L_{i,k}} = 0 \qquad \checkmark$$

Wages only redistribute b/w wage payments and tax revenues when the planner controls all prices.

Assumption A1: policy-induced changes in *foreign* relative wages have no first-order effect on home income at the optimum.

Corollary 1: $\dfrac{\partial\,\mathrm{I}_i(\cdot)}{\partial \boldsymbol{w}} = \mathbf{0} \Rightarrow$ wages drop from all FOCs

Optimal Policy: First-Order Conditions Back

FOC w.r.t. any policy instrument $\tilde{P}$ (after applying Corollary 1):

$$\sum_n \frac{\partial \mathcal{W}}{\partial W_n} \frac{\partial V_n}{\partial \tilde{P}} + \sum_n \frac{\partial \mathcal{W}}{\partial W_n} \frac{\partial V_n}{\partial E_n} \left[\frac{\partial I_n}{\partial \tilde{P}} + \frac{\partial I_n}{\partial \boldsymbol{R}} \frac{\partial \boldsymbol{R}}{\partial \tilde{P}} + \frac{\partial I_n}{\partial \boldsymbol{H}} \frac{\partial \boldsymbol{H}}{\partial \tilde{P}} + \frac{\partial I_n}{\partial \boldsymbol{C}} \frac{\partial \boldsymbol{C}}{\partial \tilde{P}} \right] = 0$$

Key derivatives (1/2):

- **Direct price effect** (Roy's id.): $\frac{\partial V_n}{\partial \tilde{P}} = -\frac{\partial V_n}{\partial E_n} C_{in,k}$ if $\tilde{P} \in \tilde{\boldsymbol{P}}_n$, else 0
- **Direct income, water:** $\frac{\partial I}{\partial \tilde{r}_{i,k}} = 0$ (water tax revenue and cost changes cancel exactly)

Optimal Policy: Key Derivatives (cont.) Back

Key derivatives (2/2):

- **Income via water stocks:** $\dfrac{\partial I_i}{\partial R_i} = -\dfrac{G_i'(\cdot)}{G_i(\cdot)} r_i \sum_k H_{i,k}$
- **Income via water use:** $\dfrac{\partial I_i}{\partial H_{i,k}} = \tilde{r}_{i,k} - r_i$
- **Policy impact on stocks:** $\dfrac{\partial R_i}{\partial \tilde{P}} = \sum_{j,k} \lambda_{ji,k} \dfrac{\partial H_{j,k}}{\partial \tilde{P}}$

Optimal Policy: Decomposition of FOCs

Back

FOCs decompose into three independent sub-problems:

$$\begin{cases} \sum_n \left(\frac{\partial \mathcal{W}}{\partial W_n} \frac{\partial \mathrm{V}_n}{\partial E_n} \pi_n \right) = \frac{\partial \mathcal{W}}{\partial W_i} \frac{\partial \mathrm{V}_i}{\partial E_i} & \text{(distributional objective)} \\ \tilde{\boldsymbol{P}} = \boldsymbol{P} & \text{(efficient non-water prices)} \\ \tilde{r}_{i,k} - r_i = \sum_n \frac{\mathrm{G}_n'}{\mathrm{G}_n} r_n \sum_k H_{n,k} \sum_g \lambda_{in,g} & \text{(efficient water prices)} \end{cases}$$

Complex GE derivatives $\partial \boldsymbol{C} / \partial \tilde{P}$, $\partial \boldsymbol{H} / \partial \tilde{P}$ cancel in both the goods and water FOCs.

Optimal Policy: Solutions Back

Solutions to the three sub-problems:

- $\pi_i^* = \dfrac{(\partial\mathcal{W}/\partial W_i)(\partial \mathrm{V}_i/\partial E_i)I_i}{\sum_n(\partial\mathcal{W}/\partial W_n)(\partial \mathrm{V}_n/\partial E_n)I_n} \Rightarrow \Delta_i^* = \pi_i^* \sum_n I_n - I_i$
- $t_{ij,k} = x_{ij,k} = s_{i,k} = 0$
- $\tau_{j,g} = \sum_i \dfrac{G_i'(\cdot)}{G_i(\cdot)} r_i \sum_k \lambda_{ji,g} H_{i,k}$